ELEC2134: Class Test 3 (Laplace Transform)

Date: Thursday 12 March 2026 (Week 4)

Time: 3 - 4 pm (Exam Duration 40 minutes).

Instruction to candidates: To maximise your marks, show all steps clearly when deriving your answers (do not write final answers only).

Pass Level Questions. Total Marks: 64

Question 1:
An RLC circuit is given below:

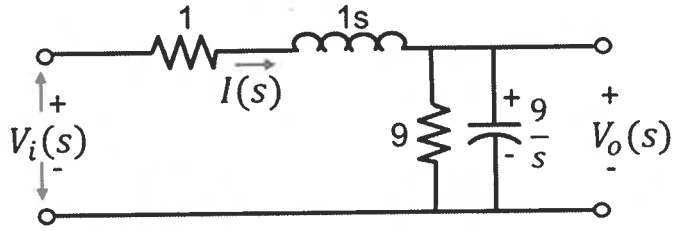
(a) Find the transfer function of the circuit. **[20 Marks]**

(b) Find the circuit response to a unit-step input. **[10 Marks]**

(c) Find the initial and final values of $v_o(t)$. **[10 Marks]**

(d) Find the impulse response $h(t)$ of the system. **[10 Marks]**

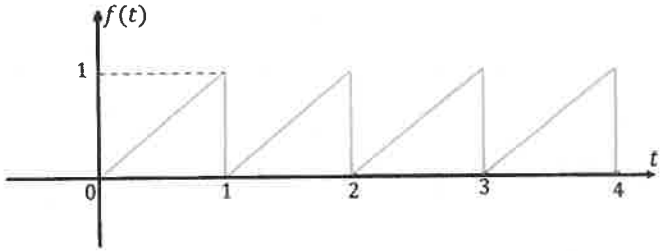
(e) Sketch $h(t)$ .Clearly indicate the key features of $h(t)$, such as its starting point, oscillatory behaviour and exponential decay.



[14 Marks]

Higher Level Questions. Total Marks: 36

Question 2:
Find the Laplace transform of the periodic sawtooth waveform defined for $t \geq 0$, as shown in the figure below.



[36 Marks]

Laplace Transform Formula Sheet for Week 4 Exam

Basic Laplace Transforms

Signal	Transform
$u(t)$	$\frac{1}{s}$
$t\,u(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$\frac{t^{n-1}}{(n-1)!}u(t)$	$\frac{1}{s^n}$
$\delta(t-\tau),\ \tau > 0$	$e^{-s\tau}$
$e^{-at}u(t)$	$\frac{1}{s+a}$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$
$\cos(\omega t)u(t)$	$\frac{s}{s^2+\omega^2}$
$\sin(\omega t)u(t)$	$\frac{\omega}{s^2+\omega^2}$
$e^{-at}\cos(\omega t)u(t)$	$\frac{s+a}{(s+a)^2+\omega^2}$
$e^{-at}\sin(\omega t)u(t)$	$\frac{\omega}{(s+a)^2+\omega^2}$

Laplace Transform Properties

Signal	Transform
$x(t)$	$X(s)$
$ax(t)+by(t)$	$aX(s)+bY(s)$
$x(t-\tau)$	$e^{-s\tau}X(s)$
$e^{a_0 t}x(t)$	$X(s-a_0)$
$x(at),\ a>0$	$\frac{1}{a}X\left(\frac{s}{a}\right)$
$x(t)*y(t)$	$X(s)Y(s)$
$-tx(t)$	$\frac{d}{ds}X(s)$
$\frac{dx(t)}{dt}$	$sX(s)-x(0)$
$\frac{d^2x(t)}{dt^2}$	$s^2X(s)-sx(0)-x'(0)$

Initial Value Theorem: $f(0)=\lim_{s\to\infty} sF(s)$

Final Value Theorem: $f(\infty)=\lim_{s\to 0} sF(s)$

If a function is periodic with Period T and $f_1(t) = f(t)[u(t)-u(t-T)]$, then the Laplace transform of $f(t)$ is : $F(s) = \frac{F_1(s)}{1-e^{-sT}}$

Class Test 3 (Laplace Transform)

Q1

(a) $$I(s) = \frac{V_i(s)}{(1+s) + \frac{9 \cdot \frac{9}{s}}{9+\frac{9}{s}}} = \frac{V_i(s)}{(1+s) + \frac{\frac{9}{s}}{\frac{s+1}{s}}}$$ — 5 Marks

$$= \frac{V_i(s) \cdot (s+1)}{(s+1)^2 + 9}$$

$$I(s) = \frac{V_i(s)\,(s+1)}{s^2+2s+10}$$ — 5 Marks

$$V_o(s) = I(s) \times \frac{9 \cdot \frac{9}{s}}{9+\frac{9}{s}} = I(s) \times 9\,\frac{1}{s+1}$$ — 5 Marks

$$V_o(s) = \frac{V_i(s)(s+1)\left[9 \cdot \frac{1}{s+1}\right]}{s^2+2s+10}$$

$$\therefore \frac{V_o(s)}{V_i(s)} = \frac{9}{s^2+2s+10}$$ — 5 Marks

(b) Let $H(s) = \frac{V_o(s)}{V_i(s)} = \frac{9}{s^2+2s+10}$

$$\therefore V_o(s) = V_i(s) \cdot \frac{9}{s^2+2s+10}$$

step input: $u(t)$ = 1 for $t \geq 0$

$$\mathcal{L}[u(t)] = \frac{1}{s}$$

$$V_o(s) = \frac{1}{s} \cdot \frac{9}{s^2+2s+10}$$ — 10 Marks

(c) Initial value: $V_o(0) = \lim_{s \to \infty} s \cdot \underbrace{\left(\frac{1}{s} \cdot \frac{9}{s^2+2s+10}\right)}_{V_o(s)} \to 0$

$$= 0$$ — 5 Marks

Final value: $V_o(\infty) = \lim_{s \to 0} s\,V_o(s)$

$$= \lim_{s \to 0} s \cdot \frac{1}{s} \cdot \frac{9}{s^2+2s+10} \quad (s \to 0,\ s \to 0,\ \tfrac{9}{10})$$

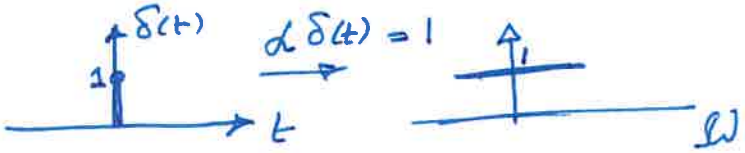
$$= 0.9$$ — 5 Marks

(d) $H(s) = \frac{V_o(s)}{V_i(s)} = \frac{9}{s^2+2s+10}$

$\therefore V_o(s) = V_i(s) \cdot \frac{9}{s^2+2s+10}$

3 Marks

When $V_i(s)$ is an impulse, $V_o(s)$ becomes impulse response.



$\therefore V_o(s) = \frac{9}{s^2+2s+10}$

impulse response: $\therefore V_o(t) = h(t)$

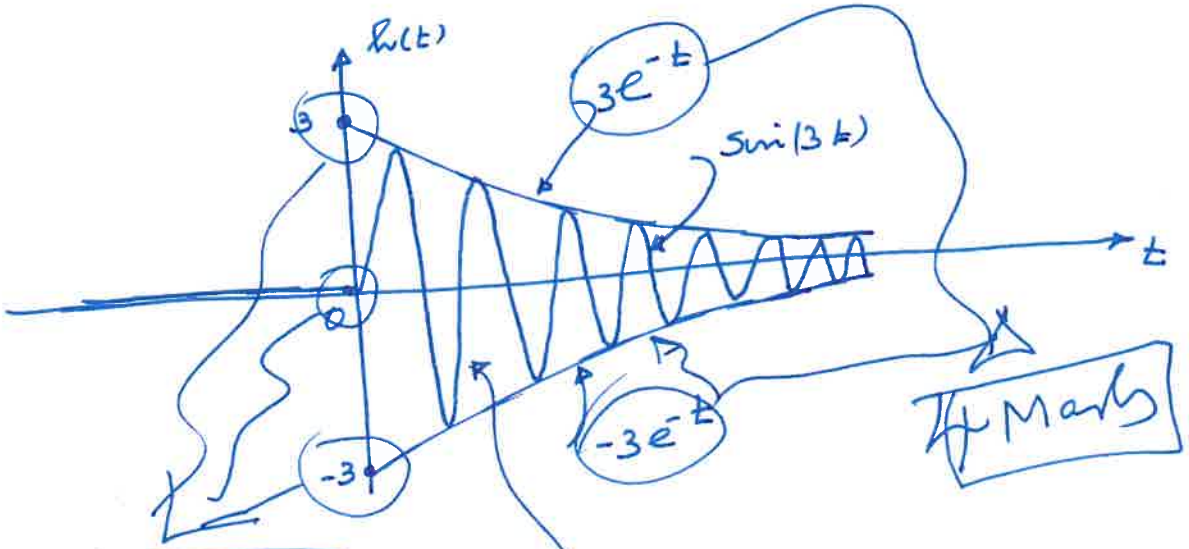
$= \mathcal{L}^{-1} \frac{9}{s^2+2s+10} = \mathcal{L}^{-1} \frac{9}{(s+1)^2-1+10}$

$= 3\mathcal{L}^{-1} \frac{3}{(s+1)^2+3^2}$

$= 3 \cdot e^{-t} \cdot \sin 3t \quad t \geq 0$

$\therefore h(t) = 3 \cdot e^{-t} \sin 3t \; u(t)$

7 Marks



5 Marks

Decaying wave

5 Marks

4 Marks

(3)

Q_2: Period $T = 1$; $f(t) = mt$ ← slope $= 1$ $\therefore f(t) = t$ } 5 Marks

$$f_1(t) = f(t)\,[u(t) - u(t-1)] \quad \leftarrow \text{5 Marks}$$

$$= t\,u(t) - t\,u(t-1)$$

$$= t\,u(t) - ((t-1)+1)\,u(t-1)$$

$$f_1(t) = t\,u(t) - (t-1)u(t-1) - u(t-1) \quad \} \text{10 Marks}$$

$$F_1(s) = \frac{1}{s^2} - \frac{e^{-s}}{s^2} - \frac{e^{-s}}{s}$$

$$= \frac{1}{s^2}\left[1 - e^{-s} - s e^{-s}\right] \quad \} \text{10 Marks}$$

$$F(s) = \frac{F_1(s)}{1 - e^{-s\cdot 1}} = \frac{1}{s^2}\,\frac{1 - e^{-s} - s e^{-s}}{1 - e^{-s}} \quad \} \text{6 Marks}$$

$$\boxed{F(s) = \frac{1}{s^2}\,\frac{1 - e^{-s} - s e^{-s}}{1 - e^{-s}}}$$